

Table 1. **Capability deep-dive:** the benchmarks that probe each of the β capability cuts, and whether each builds a native VLA-vs-world-model contrast. Native contrasts cluster in the hidden-state, theory-of-mind and counterfactual *reasoning* cuts (LIBERO-Mem, ComPhy, AGENT, BIB, ACRE, CoPhy, CausalWorld); the long-horizon and social *task-success* cuts are largely model-agnostic (ALFWorld’s abstract-to-grounded transfer is the lone partial). **Eval:** *open*/*closed*. **VLA vs WM:** ✓ native contrast, ~ partial, -- model-agnostic; each cut’s header tallies the fraction of its benchmarks that build any (native or partial) contrast.

Benchmark	Year	Eval	What it probes	VLA vs WM
Hidden-state / memory / occlusion				3/4
LIBERO-Mem [9]	2025	<i>closed</i>	occluded object state carried across time	✓
TEAch [17]	2021	<i>closed</i>	partner belief and history via dialog	--
ComPhy [8]	2022	<i>open</i>	hidden mass and charge inferred from collisions	✓
Bongard-HOI [11]	2022	<i>open</i>	few-shot hidden-concept induction	~
Theory-of-mind				3/4
AGENT [22]	2021	<i>open</i>	expectations over an agent’s goals and actions	✓
BIB [10]	2021	<i>open</i>	infant-style goal inference from behaviour	✓
PHASE [16]	2021	<i>closed</i>	social relations and goals inferred from motion	~
SocialAI [12]	2021	<i>closed</i>	language-based social interaction with agents	--
Counterfactual / physical inference				3/6
WorldPrediction [7]	2025	<i>open</i>	discriminate correct vs counterfactual action	--
Physion [3]	2021	<i>open</i>	predict a physical outcome from vision	--
Physion++ [23]	2023	<i>open</i>	infer latent physical properties online	--
ACRE [25]	2021	<i>open</i>	abstract causal reasoning over blicket tests	✓
CoPhy [2]	2019	<i>open</i>	counterfactual outcome under a changed cause	✓
CausalWorld [1]	2020	<i>closed</i>	causal do-interventions on task variables	✓
Long-horizon / compositional				1/6
CALVIN [15]	2021	<i>closed</i>	chain several language subtasks in a row	--
BEHAVIOR-1K [14]	2024	<i>closed</i>	1000 long-horizon household activities	--
ALFRED [20]	2019	<i>closed</i>	instruction to action under partial view	--
ALFWorld [21]	2020	<i>closed</i>	abstract-text to grounded-task transfer	~
EmbodiedBench [24]	2025	<i>closed</i>	broad long-horizon embodied planning	--
PARTNR [5]	2024	<i>closed</i>	human-robot collaborative household tasks	--
Social / multi-agent				0/5
Habitat 3.0 [18]	2023	<i>closed</i>	human-robot co-habitation and hand-off	--
SocNavBench [4]	2021	<i>closed</i>	socially compliant navigation	--
Melting Pot [13]	2021	<i>closed</i>	mixed-motive multi-agent generalization	--
SMAC [19]	2019	<i>closed</i>	cooperative multi-agent micro-control	--
CrowdNav [6]	2018	<i>closed</i>	socially-aware navigation among pedestrians	--